



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SAF FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

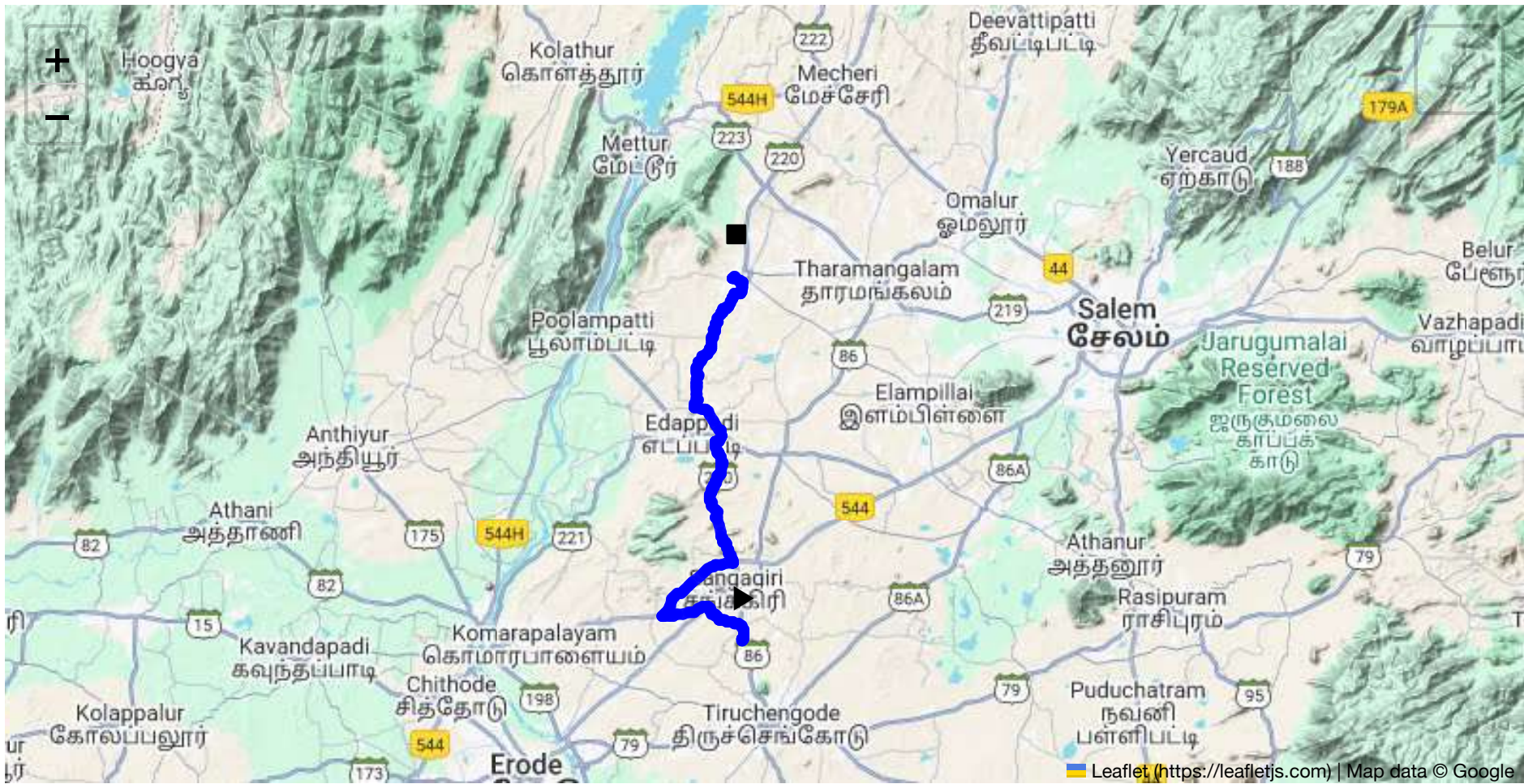
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 42.99 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours

Start: (11.4381, 77.8734)
End: (11.69643, 77.868404)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Idupali via Padaiveedu spans approximately 42.99 kilometers. This route primarily travels along rural roads and highways, combining state highways and regional roads typical of Tamil Nadu, offering direct access through the district's agricultural and semi-urban areas.

Weather Conditions

Tamil Nadu experiences a tropical climate, with significant variations based on the season:

- **Summer (March to June):** High temperatures (up to 40°C) can affect vehicle performance.
- **Monsoon (July to September):** Heavy rain leading to potential flooding and reduced visibility.
- **Winter (November to February):** Mild and typically dry, presenting fewer weather-related hazards.

Drivers should prepare for potential flooding or reduced road grip during the monsoon, especially near low-lying areas or river crossings.

Traffic Patterns

Traffic is typically moderate to heavy on Bhavani Main Road, especially during:

- **Morning (8-10 AM) and Evening (5-7 PM):** Peak hours due to commuter and commercial traffic.
- Market and festival days can also increase congestion significantly.

Expect higher congestion near any larger towns or intersections along the Bhavani Main Road.

Road Quality and Infrastructure

Mixed road quality can be expected:

- **Main Highways:** Generally well-maintained, but potholes and rough patches may be present.

- **Secondary Roads:** Narrower with inconsistent pavement. Expect occasional roadwork or diversions.

Drivers should be cautious in rural areas where road conditions may vary unexpectedly.

Alternative Routes

In case of diversions or emergencies, an alternative route via smaller roads can be explored:

- Heading east towards Namakkal then south to Idupali, though longer, can avoid congestion on the main highways.
- Google Maps or local navigation apps should be consulted real-time for updates.

Local Regulations on Hazardous Materials

Tamil Nadu has specific restrictions on transporting hazardous materials through densely populated areas and during night hours. Permits are required, and adherence to safety protocols and signage is mandatory.

Historical Incidents

There have been occasional accidents involving heavy vehicles in the region, often due to sharp turns, ineffectual braking on slopes, or poor visibility during the monsoon. These incidents highlight the need for caution and adherence to speed limits and vehicle checks.

Environmental Considerations

The route passes through regions with agricultural activity. Care should be taken to prevent contamination from spillage. Additionally, there may be protected natural areas or water bodies nearby, requiring drivers to avoid potential contamination.

Communication Coverage

Mobile network coverage varies:

- **Main Highways and Urban Centers:** Generally full coverage.
- **Rural Areas and Woodland:** Occasional dead zones, which could affect GPS reliability and emergency communication. Pre-loading maps and having alternative navigation options is advisable.

Emergency Response Times

- **Urban Areas:** Typically faster response times (within 30 minutes).
- **Rural Sections:** Could experience delays (up to 1 hour) due to isolated locations and access difficulties.

Overall Risk Assessment

This route presents moderate risk, with traffic congestion, varying road conditions, and communication dead zones being significant factors. Proper preparation—such as obtaining necessary transport permits, allowing for weather conditions, and using reliable communication tools—is essential for mitigating risks effectively. Regular updates on traffic and road conditions should be gathered to navigate efficiently and safely.

Truck drivers should be well aware of local laws regarding hazardous materials and be trained in emergency procedures to ensure safety and compliance throughout the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45358, 77.85781	30 KM/Hr
7	Turn	Medium	11.45353, 77.85697	30 KM/Hr
8	Turn	Medium	11.45839, 77.85143	30 KM/Hr
9	Turn	Medium	11.46053, 77.85034	30 KM/Hr
10	Turn	Medium	11.46307, 77.84941	30 KM/Hr
11	Turn	Medium	11.46316, 77.84921	30 KM/Hr
12	Turn	High	11.45542, 77.81725	15 KM/Hr
13	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
14	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
15	Turn	Medium	11.50885, 77.86037	30 KM/Hr
16	Turn	Medium	11.52571, 77.85522	30 KM/Hr
17	Turn	Medium	11.52744, 77.85582	30 KM/Hr
18	Turn	Medium	11.55460, 77.85624	30 KM/Hr
19	Turn	High	11.57893, 77.85430	15 KM/Hr
20	Turn	Medium	11.57903, 77.85441	30 KM/Hr
21	Turn	Medium	11.60978, 77.84165	30 KM/Hr
22	Turn	Medium	11.61795, 77.84020	30 KM/Hr
23	Turn	Medium	11.64055, 77.84600	30 KM/Hr
24	Turn	Medium	11.64173, 77.84878	30 KM/Hr
25	Turn	Medium	11.65486, 77.85335	30 KM/Hr
26	Turn	Medium	11.67310, 77.86031	30 KM/Hr
27	Turn	Medium	11.68196, 77.86690	30 KM/Hr
28	Turn	High	11.68492, 77.86898	15 KM/Hr
29	Turn	Medium	11.68463, 77.87262	30 KM/Hr
30	Turn	Medium	11.68486, 77.87288	30 KM/Hr
31	Turn	High	11.69287, 77.87436	15 KM/Hr
32	Turn	Medium	11.69316, 77.87208	30 KM/Hr
33	Turn	Medium	11.69339, 77.87196	30 KM/Hr
34	Turn	Medium	11.69500, 77.86995	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
35	Turn	Medium	11.69513, 77.86800	30 KM/Hr
36	Turn	Medium	11.69560, 77.86728	30 KM/Hr
37	Turn	High	11.69576, 77.86724	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



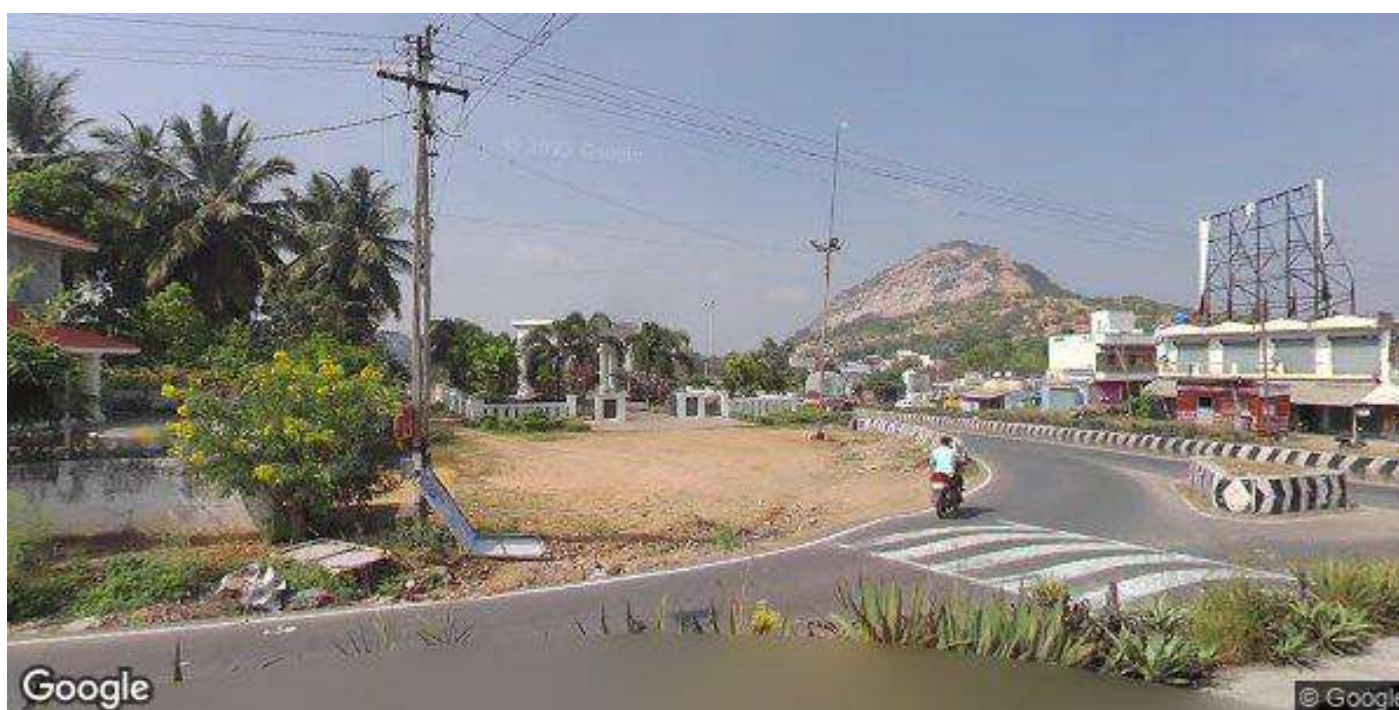
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46053, 77.85034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49390, 77.86748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50885, 77.86037



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52571, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52744, 77.85582



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



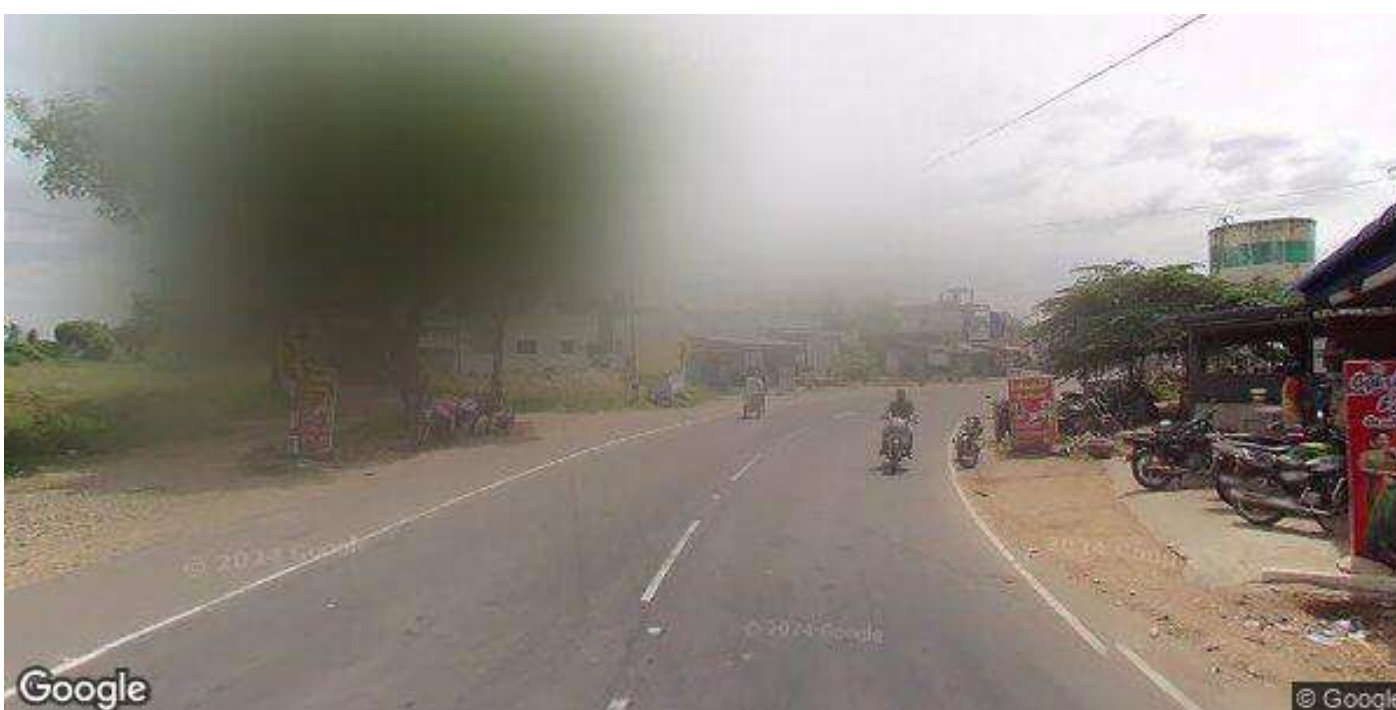
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60978, 77.84165



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61795, 77.84020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64055, 77.84600



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64173, 77.84878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65486, 77.85335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67310, 77.86031



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68196, 77.86690



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68492, 77.86898



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68463, 77.87262



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68486, 77.87288



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69287, 77.87436



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69316, 77.87208



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69339, 77.87196



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69500, 77.86995



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69513, 77.86800



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69560, 77.86728



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69576, 77.86724

